

Seeds, their structure and germination

Have you ever observed the seeds in a fruit or a vegetable? We all come across seeds when we eat different kinds of fruits and vegetables. Some seeds are small, others are large. Have you ever grown some seeds in your farm, flower pot or home garden? A seed grows into a baby plant and then a tall tree. What does a seed need to grow into an adult plant? Do the seeds eat? Where do they get the energy to grow? Are the seeds of all plants similar? Is there any difference between the seeds of different plants?



Figure 3.1 Fruits with seeds

Do you recognize these fruits and their seeds? Which of the seeds are edible? Which of these are not edible?

Explore

Are there any fruits without seeds? **Name a fruit without seed.**

In this chapter, you will learn about:

- The structure and germination of a French bean seed.
- The structure and germination of a maize seed.
- Function of Cotyledons
- Conditions necessary for germination.

All the students will be able to:

- ✓ Compare the structure and function of a French bean seed and a maize seed.
- ✓ List the functions of Cotyledons.
- ✓ Identify the conditions necessary for germination.
- ✓ Predict what would happen to a plant if conditions necessary for germination are not fulfilled.
- ✓ Conduct an investigation to assess your prediction.

Structure and germination of French bean seed

Have you ever eaten French beans? You know that seeds are parts of the plant that grow into new plants. Observe the picture; it shows a bean pod. It is a fruit, and there are several seeds or beans inside it.



Figure 3.2: French beans



Figure 3.3: French bean pod

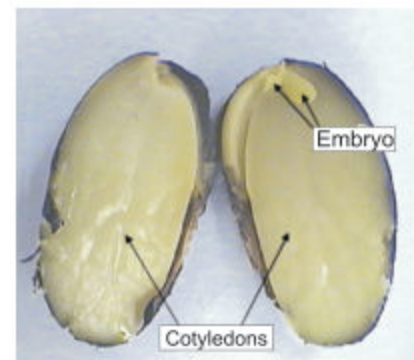


Figure 3.4: Transection of a French bean

Parts of a French bean seed:

Seed coat: Each seed is covered on the outside by a seed coat. The seed coat protects the seed. The scientists call the seed coat a **testa**. There is a small opening in the seed called **micropyle**.

Cotyledons: Soak a bean in water, remove its seed coat, press it gently between your fingers, it will split into two halves. Each half is a Cotyledon. The plants which have these kinds of seeds are called **dicotyledonous** (di; two) or **dicots**. A French bean plant is a dicot.

Embryo: Inside the split French bean seed you will see a mini plant. This is called the **embryo**. Embryo is the part of the seed that develops into the new plant. It contains structures that develop into the root, stem and leaves of the new plant.

Unscramble the
given word:
ASETT

Hint: The scientific
term for a seed coat.

Keep in mind that French
beans are dicotyledonous!



Seed
talk!

Germination of a French bean seed

Do you know what does germination mean? Germination or seed germination is a process by which seeds sprout and begin to grow. When a French bean seed germinates in soil in the presence of water and oxygen, the embryo starts developing.



Germination of a French bean

The root emerges first. It grows downwards.

Then a hook like structure emerges from the soil. This hook like structure is called a **hypocotyl**. The **hypocotyl** pulls the **cotyledons** containing the **embryo** out of the soil.

The cotyledons that are now above the soil form the tip of the plant.

Then the seed coat falls to the soil.

After sometime the leaves of the plant appear. They start **photosynthesis**.

The cotyledons shrivel and fall off the plant.



Figure 3.5: Formation of hypocotyle

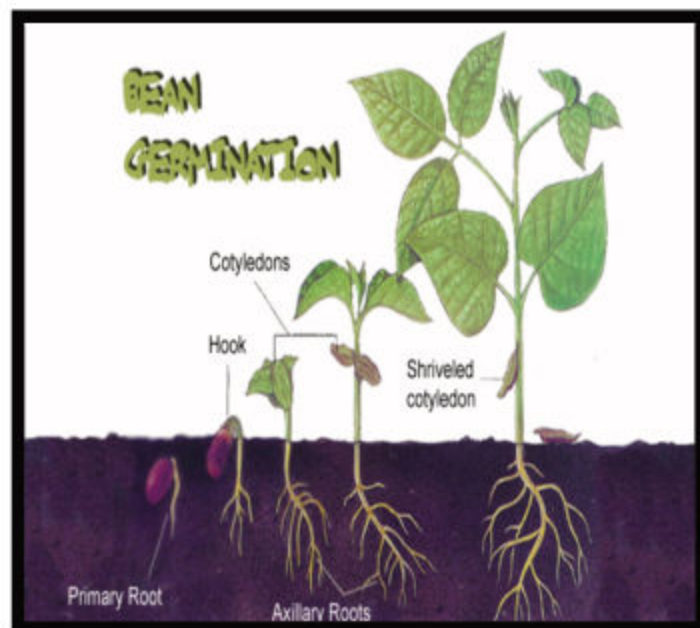


Figure 3.6: Germination of seed

Do you know that this type of seed germination in which the cotyledons emerge above the ground is called **epigeal germination**?



Figure 3.7: Chickpeas



Do you know that the pea and chickpea also germinate just like French beans?

Unscramble the given word:
leeaigp

Hint: The kind of germination in which cotyledons emerge above the ground.

Do you know that there are thousands of different kinds of beans. They all germinate the same way.



Figure 3.8: Different kinds of beans

Structure and germination of a maize seed

This is a maize fruit with many seeds. You must have eaten it as roasted seeds or a roti of maize seeds.

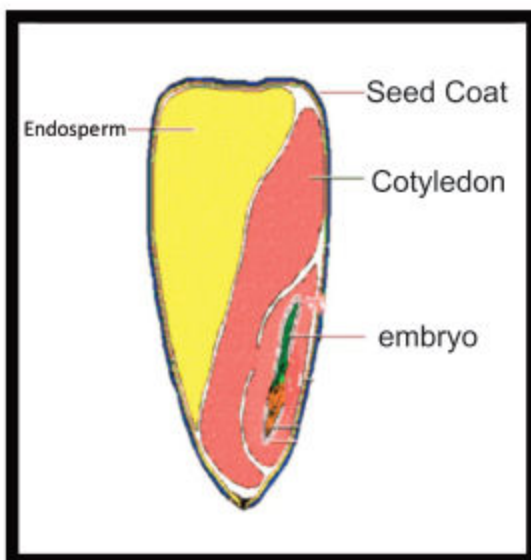


Figure 3.9: Transection of a maize seed

Parts of a maize seed:

Seed coat: The maize seed is surrounded by a seed coat. It has an opening called **micropyle**. If the teacher cuts a maize seed longitudinally, using a razor blade, you will see the following structures:

Embryo: A part of the seed that develops into a new plant. It has structures that develop into the root, stem and leaves of new plant.

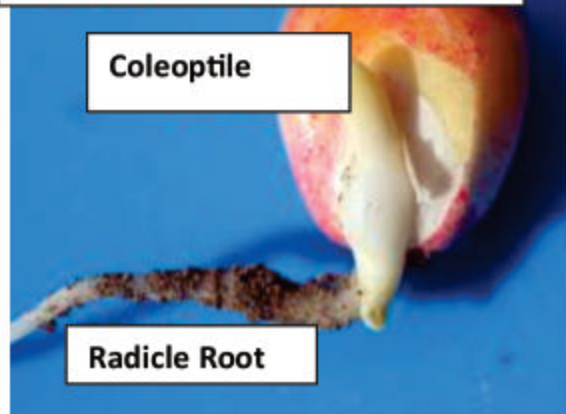
Endosperm: A part of the seed that contains stored food. The stored food is used by embryo when it germinates.

Cotyledon: Since maize is a **monocot**, there is a single cotyledon in a maize seed. **Cotyledons** supply food from the endosperm to the developing embryo.

Germination of maize seed

You know that germination is a process by which seeds sprout and grow into seedlings. When a maize seed is sown in soil, the embryo starts developing.

A seedling's appearance 60 hours after plantation



Germination of a maize seed

The seed develops a root first. The root starts growing downwards.

A straight stalk grows out of the soil. This stalk like structure is called a **coleoptiles**. Then it splits and the new stem and leaves of the plant come out and grow upwards.

The **cotyledons** and **seed coat** remain below the soil.

During germination, the seed gets its food from the stored food in the **endosperm**. **Photosynthesis** does not take place until the first leaves develop.

This kind of germination in which cotyledons remain inside the soil is called **hypogeal germination**.



Unscramble the given word:
oopleilte

Hint: The stalk like structure that comes out of soil during germination of maize

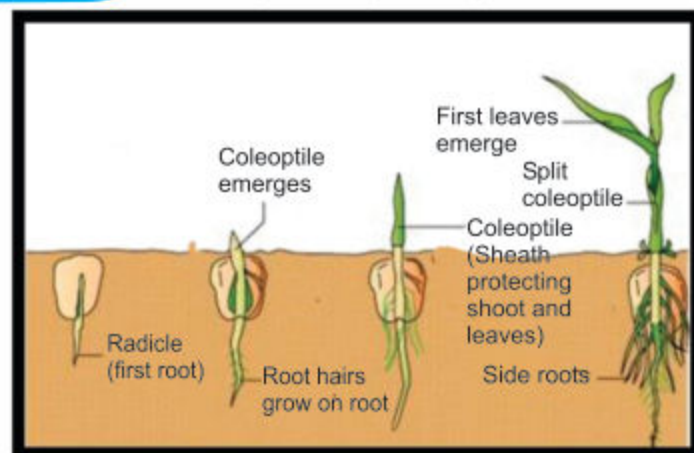


Figure 3.10: Germination of a maize seed

Activity 1: Studying the structure of french bean seed and maize seed and watching them germinate.

What do I need?

Dry and pre-soaked bean seeds and maize seeds, hand lens, blade, transparent glasses, tissue paper and water

What to do:

1. Observe a dry bean seed. Locate the seed coat and micropyle.
2. Take a pre-soaked bean seed. Split open a pre-soaked bean. Locate embryo and cotyledons. Draw what you see.
3. Similarly observe a dry maize seed. Locate seed coat and micropyle. Take a pre-soaked maize seed. Cut a pre-soaked maize seed longitudinally into two halves. Locate embryo, endosperm and cotyledons. Draw what you see.
4. With the help of your teacher set up two separate glass jars for germination of the seeds. Spread a cloth piece or tissue paper along the walls of the jars and put a little water at its base.
5. Slip seeds into the space between the glass wall and the cloth/tissue paper. Make sure they are kept in various directions as shown in the picture.
6. Watch your seedling glass jars every day. Make sure that the water is one inch deep. But it should not reach the seeds.
7. Record your observations in the table every day.



Figure 3.11: Bean seeds planted in glass jars

What I observed:

Copy this table in your science notebook. Extend it up to 20 days. Observe your set up everyday till 20 days. Write or draw in the table. Take photographs if possible.

Days	What's new in the bean jar today?	What's new in the maize jar today?

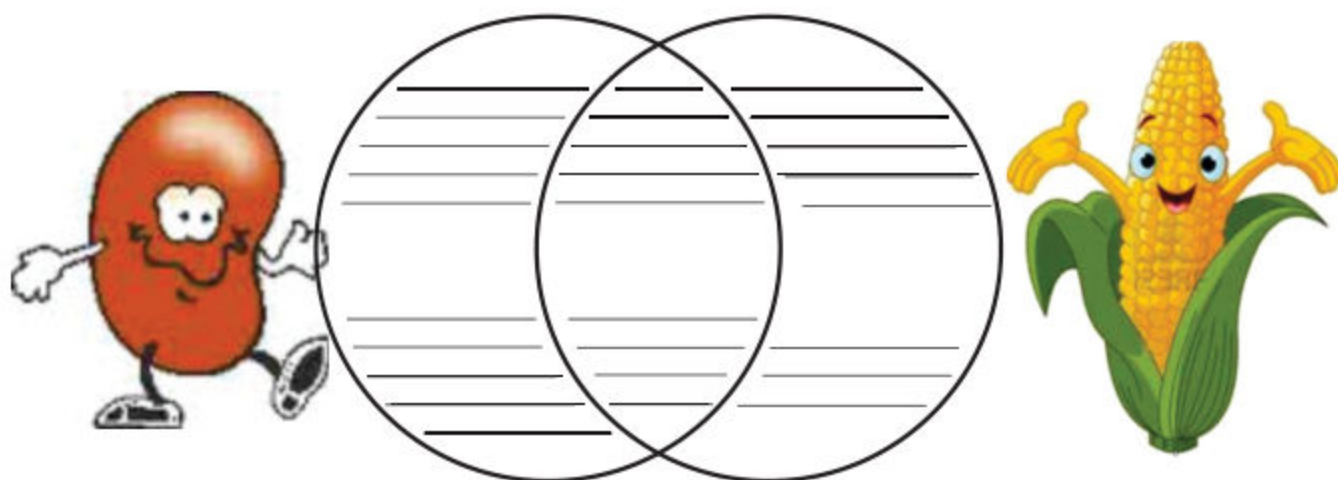
Teacher Note: Arrange dry and pre-soaked seeds and show each structure to the students while referring to the diagram and explanation given in the textbook. Draw their attention to the changes taking place during germination.

Do you see how tiny seeds develop into beautiful plants which benefit us? Do you know how tiny seeds are protected during their germination and growth. Talk to your teacher and classfellows about the process.

Compare the structure and germination of a French bean and a maize seed

Comparison of structure and germination of a French bean seed and a maize seed

Do you see that there are some important differences in the structure as well as in the germination of French bean and maize seeds? We have found that a maize seed has one cotyledon while a French bean has two. Did you notice some other similarities and differences in the structure and germination of the two seeds? Fill up this Venn diagram.



French bean and maize plants are showing each other their childhood photographs.

Childhood photographs

This one is from the day when I grew my root.



And the day when the hook formed.

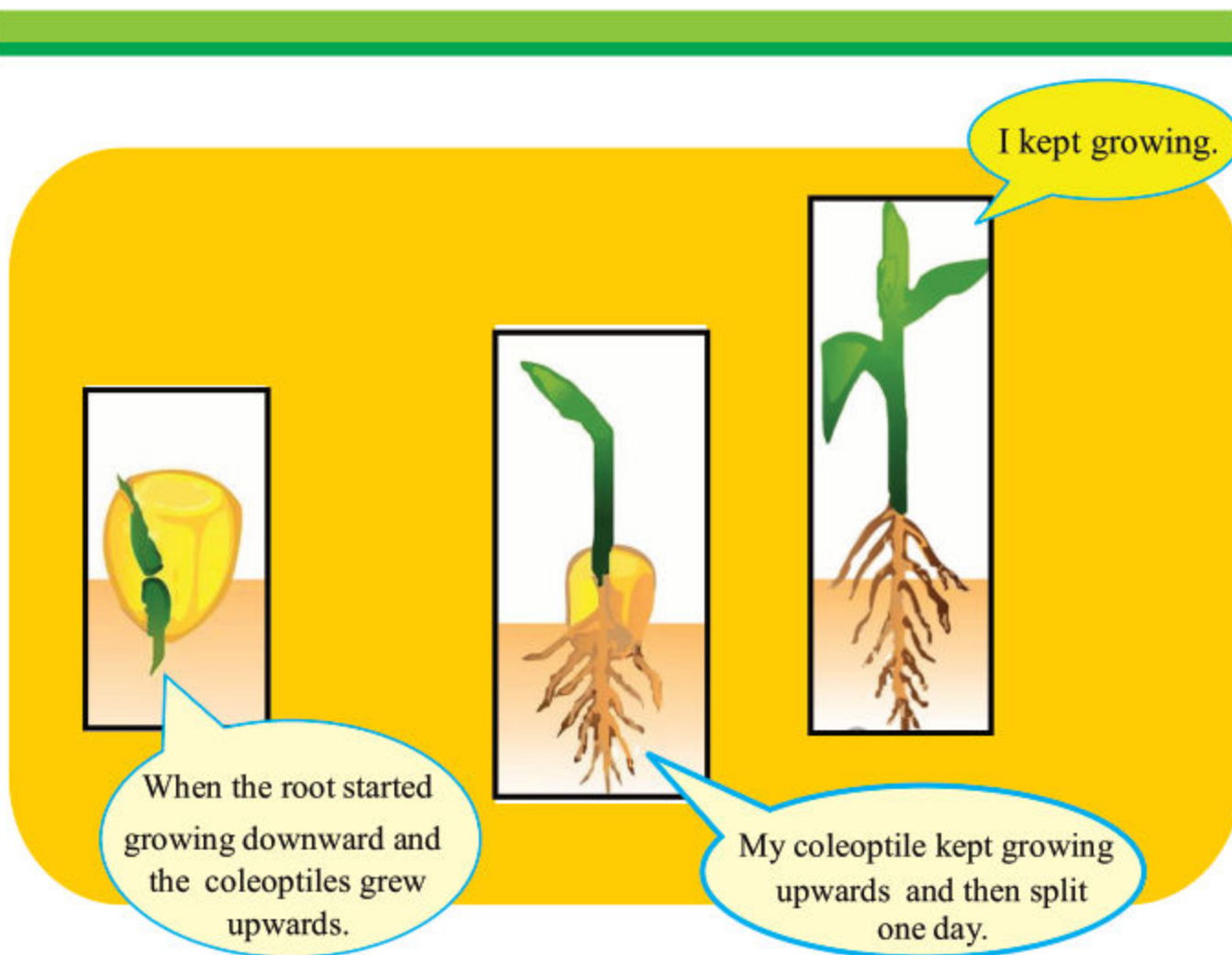


This is when the cotyledons turned green.



Finally when the leaves appeared and the cotyledons shrivelled up.





Function of cotyledons

 List the function of cotyledons.

You know that cotyledons are a part of the seed. Do you know what cotyledons do? They play an important part in seed germination.

In a mature seed, cotyledons contain food themselves or can get food from the endosperm, for the developing seedling. Thus they provide the energy necessary for germination.

Unscramble the given word:
yledontoc

Hint: The part of the seed that has access to stored food or stores food itself.

In one kind of germination like that of the French bean seed, the **cotyledons** grow out of the soil. They become **photosynthetic**. This means that they take in water and absorb sunlight to make food for the plant. Thus, they act as the first leaves of the seedling. In many cases, they stay on the plant for quite some time after the first leaves appear.



Do you know humans also consume the food stored in the cotyledons of seeds such as peanuts and chickpea. example of a seed that we use for food.



Unscramble the given word:
naiontrmeg

HINT: The sprouting of a seed into a seedling.

The conditions necessary for germination

Identify the conditions necessary for germination.

Millions of seeds are produced every day by various plants. Do all these seeds germinate? Which of these seeds are able to germinate and grow into new plants?

Which of the seeds fail to germinate?

Only the seeds that receive the right conditions for germination are able to germinate. What are these conditions? Scientific experiments have shown that following conditions must be fulfilled for a seed to germinate.

Oxygen: It is necessary for life. For the seed to be able to germinate it must receive oxygen. When the seeds receive oxygen they start respiration. Respiration breaks down the stored food in seed to give it energy for growth.



Water: Seeds need water to swell them up in order to break the seed coat. Without water, the seed coat of a dry seed cannot break itself open.

Proper temperature: Every kind of seed germinates at a particular range of temperature. Most seeds will germinate between 16-24°C; however, there are some seed kinds which need colder temperatures i.e., 2-4°C, such as radish and spinach.



Some seeds also need exposure to light for their germination. While some other seeds require complete darkness.

And what about sunlight?

You know they must get water, oxygen and warmth.



Figure 3.12: Children planting sprouts

 Predict what would happen to the plant if the conditions necessary for germination are not fulfilled. Conduct an investigation to assess your prediction.

Do you think your seeds might be able germinate if we don't give them oxygen, water and warm temperature? Most probably no! Let us test our idea using an experiment.

Activity 2: Investigating whether a seed will germinate in the absence of oxygen, water or in refrigerator temperature.

What do you know? Seeds are the parts of plants that grow into new plants. All seeds do not germinate in order to produce new plants.

What do you want to know from your experiment?

I want to know whether a French bean seed will germinate in the absence of water, air and proper temperature.

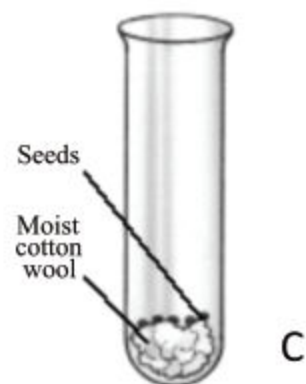
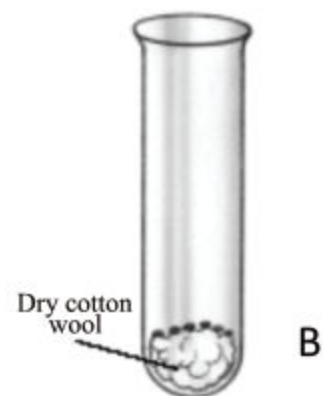
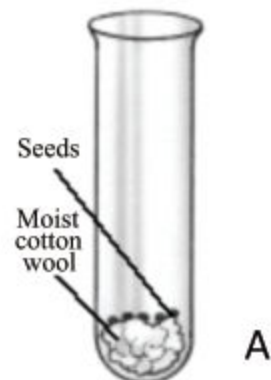
What do you need?

French bean seeds, four glass jars or test tubes, cotton wool, boiled and cooled water, and a refrigerator.

What to do:

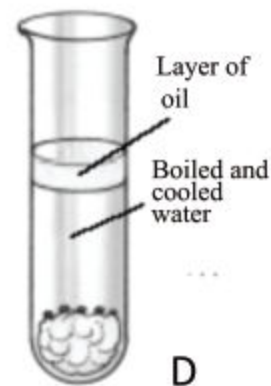
I will take four sets of seeds.

- I will germinate the first group of seeds (A) in moist cotton wool (in the presence of water and air at room temperature).
- I will germinate the second set of seeds (B) in dry cotton wool (in the absence of water at room temperature).
- I will germinate the third set of seeds (C) in moist cotton wool in refrigerator (in the presence of water and air at low temperature).
- I will germinate the fourth set of seeds (D) in boiled cooled water with a layer of oil above it (in the presence of water and absence of air).



What do you think will happen? (Use the words **will** or **will not**, to complete the sentences.)

- Seeds _____ germinate in the first jar in the presence of water, air and proper temperature.
- Seeds _____ germinate in the second jar in the absence of water.
- Seeds _____ germinate in the third jar in the absence of proper temperature
- Seeds _____ germinate in the fourth jar in absence of air.



Jar / Test tube	What happened to the seeds? Did the seeds begin germination?
Jar 1 (with water, air and proper temperature)	
Jar 2 (without water)	
Jar 3 (without proper temperature)	
Jar 4 (without oxygen)	

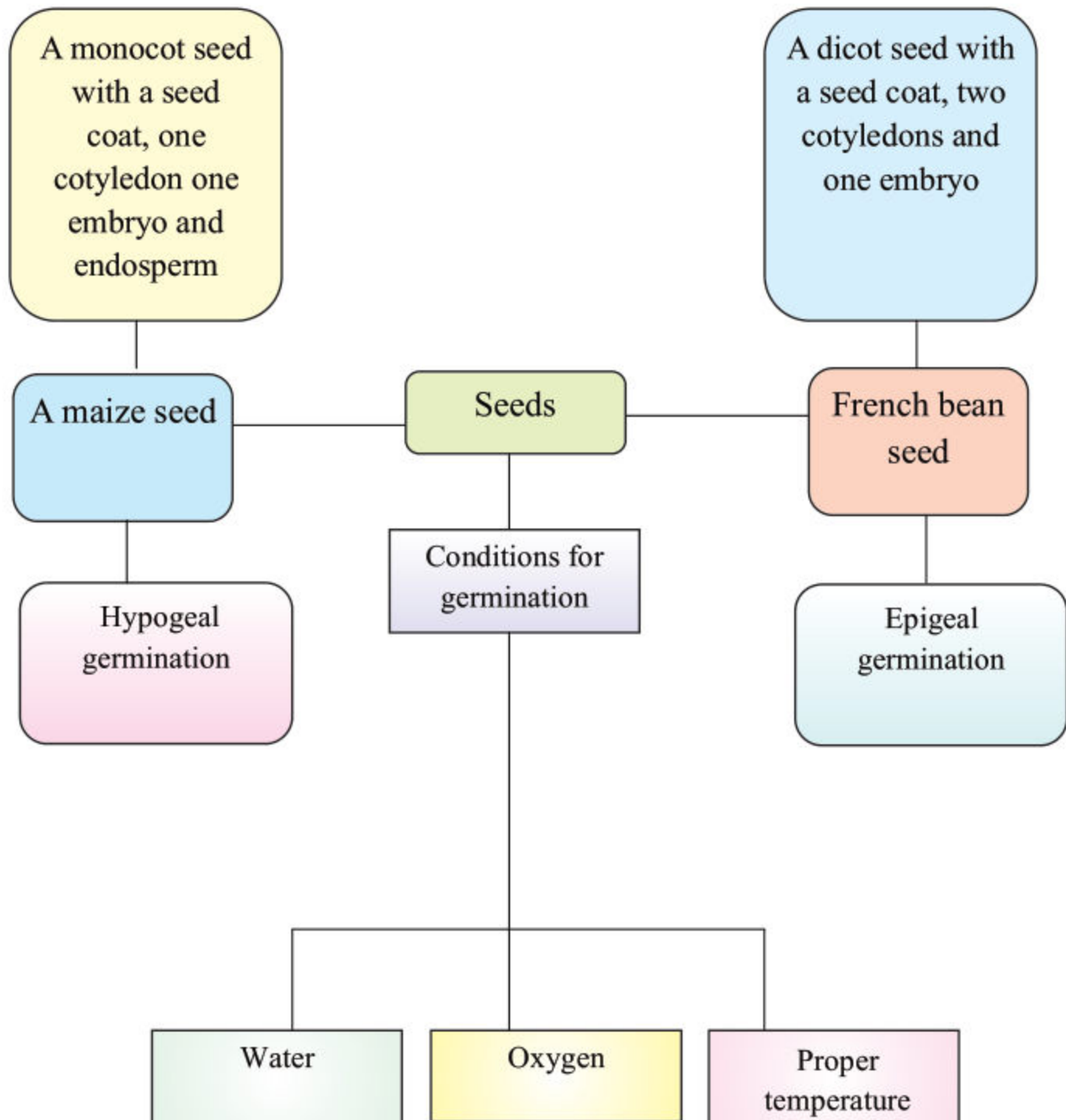
What happened in your experiment?

The experiment shows that

I conclude that:

Teacher Note: Engage the students in performing the experiment in groups or demonstrate it. Arrive at a conclusion along with the student.

Summary



Project

“Better seeds, better crops, better food”

What I need:

- List of questions to explore how seed germination is important in order to deal with the world’s food problems.
- List of people/organizations that you could talk to, consult and visit.
- List of books and websites.
- Poster/ Chart paper and markers.

What to do:

1. Explore these questions. Add some more to this list.
 - Have the scientists developed any techniques to test the quality of a seed before it germinates?
 - What techniques are being used today to achieve better germination of seeds of crop plants? Are there any chemicals or machines that help achieve better germination?
 - Do pesticides affect seed germination?
2. Discuss with your family members, teachers, pair and senior fellows to find answers to the given questions and those that you have listed.
3. Explore from books, websites, tv and radio or text your family and friends.
4. Present your findings through a poster/Power Point presentation, story or role play entitled “Better seeds; Better crops; Better food”.

Review questions

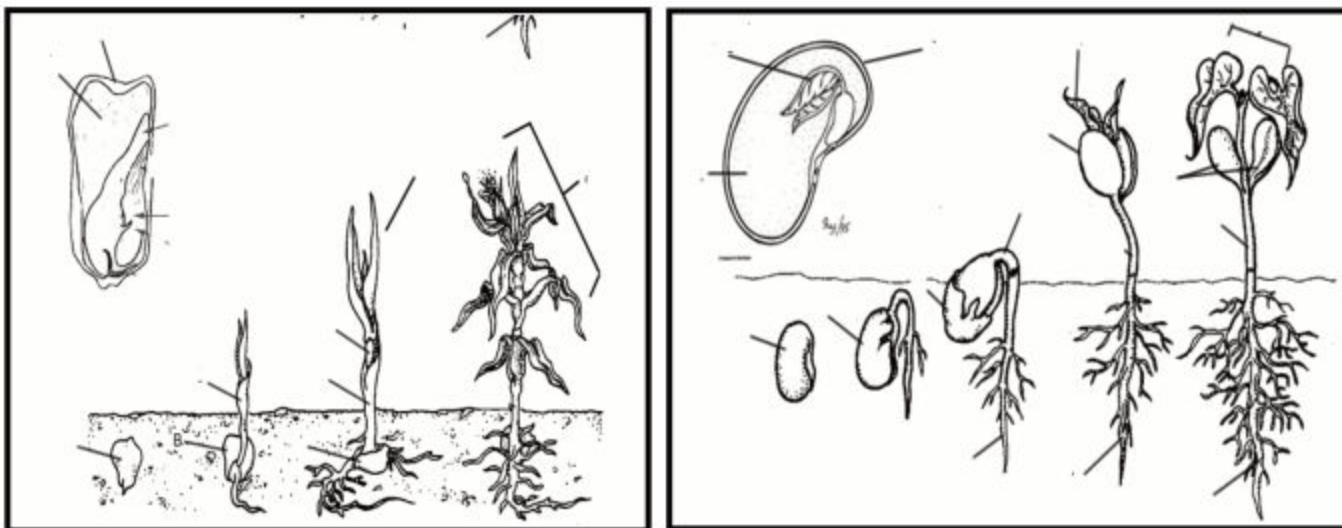
1. Circle T for True and F for False statements :

- a) A French bean seed has one cotyledon. T F
- b) A maize seed has two cotyledons. T F
- c) In the germination of a French bean seed, the cotyledons come above the ground and perform photosynthesis. T F
- d) Water, air and proper temperature are the conditions required for germination. T F
- e) In the germination of a maize seed, the cotyledons stay below the soil. T F

2. Choose the correct answer

- (i) The germination in which the cotyledons come above the ground.
 - a. Hypogeal germination
 - b. Epigeal germination
 - c. Germination
- (ii) These conditions required for germination of a seed:
 - a. Carbon-dioxide, water, proper temperature
 - b. Oxygen, dry soil, proper temperature
 - c. Oxygen, proper temperature, water
- (iii) About the requirement of light for germination.
 - a. Some seeds germinate in dark whereas some require light.
 - b. All seeds germinate in dark.
 - c. All seeds germinate in light.
- (iv) Hypocotyls hook is formed during the germination of:
 - a. Maize seed
 - b. Bean seed
 - c. All seeds
- (v) Coleoptile is a structure found in:
 - a. Maize seed
 - b. Bean seed
 - c. All seeds

3. Colour and label the diagrams given below:



4. Collect the following fruits. How many seeds do you find per fruit? Look at their structure. You may pre-soak them to have a better look. Then fill in the following table.

Fruits	How many seeds do you find per fruit?	No. of cotyledons	Do you see an embryo inside the seed?
Melon or Watermelon			
Lemon			
Green chillies			
Peach			
Pea Pod			

Paste the seeds in your notebook and write the important information about them

5. Answer the following questions

- Differentiate between epigeal and hypogeal germination.
- Describe the conditions required for germination of seeds.
- What are the functions of cotyledons?

6. Scientific Problem Solving

Robina is trying to germinate sunflower seeds. It is summer. She put the seeds in a glass jar with moist cotton wool and places the set up in hot sunny place. A week has passed and the seeds have not germinated. Now Robina is thinking what might have stopped the seeds from germinating. Can you help her sort out the problem? What could be some reasons?

